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FACULTY OF CYBER PHYSICAL SYSTEM ENGINEERING

LAB REPORT 02

BASIC 2D DRAWING WITH TURTLE GRAPHICS

COURSE CODE: IRE 404

COURSE TITLE: COMPUTER GRAPHICS AND ANIMATION SESSIONAL

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Task 1: Drawing a Smiling Face

GitHub Repository Link:
https://github.com/Masud744/Computer-Graphics-Lab/blob/main/task1.py

Python Source Code (`task1.py`):

```python
from turtle import *

speed(3)

# 1. Main face outline (radius 300 circle)
pensize(3)
pencolor("black")
penup()
goto(0, -300)
pendown()
circle(300)

# 2. Smiling mouth
# Red blot first, then a white blot slightly above it
    to make a crescent
penup()
goto(0, -90)
pendown()
dot(180, "red")

penup()
goto(0, -65)
pendown()
dot(180, "white")

# 3. Nose (thick blue triangle)
# Drawn after mouth so the white blot doesn't cover it
pensize(5)
pencolor("blue")
penup()
goto(0, 30)
pendown()
goto(-35, -40)
goto(35, -40)
goto(0, 30)

# 4. Eyes (small green filled circles)
penup()
goto(-90, 90)
pendown()
dot(40, "green")

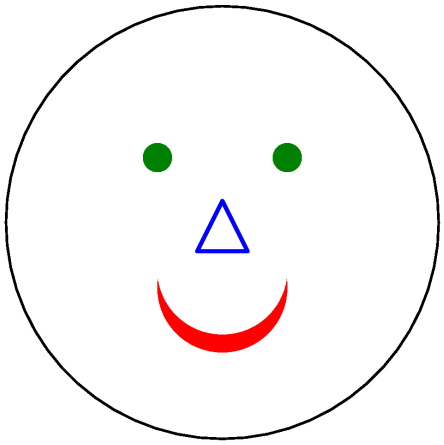
penup()
goto(90, 90)
pendown()
dot(40, "green")

hideturtle()
done()
```

Problem Description: Draw a face enclosed in a black circle (radius 300), small green blots for eyes, a thick blue triangle for a nose, and a red smiling mouth. The smiling mouth is synthesized by drawing a red blot, and then overlaying a slightly higher white blot to leave a red crescent.

Methodology: The face boundary is drawn with `circle(300)` from $(0, -300)$. The crescent smile is formed via color occlusion: a red disk (`dot(180)`) at $(0, -90)$ is overlaid with an offset white disk at $(0, -65)$. The mouth is drawn before the triangular nose and green eyes so the white blot does not overwrite foreground features.

Output Visualization:



Result & Observation: The mouth must be rendered prior to the nose and eyes. Proper Z-ordering ensures the crescent smile is cleanly carved out without overlapping the foreground nasal and eye contours.

Task 2: Flag Rectangle and Proportions (10:6)

GitHub Repository Link:
https://github.com/Masud744/Computer-Graphics-Lab/blob/main/task2.py

Python Source Code (`task2.py`):

```python
from turtle import *

speed(3)

# Flag dimensions with 10:6 ratio
length = 600
height = 360

# 1. Green background rectangle
penup()
goto(-300, -180)
pendown()
color("green", "green")
begin_fill()
for i in range(2):
    forward(length)
    left(90)
    forward(height)
    left(90)
end_fill()

# 2. Red circle inside
# Circle centre is placed at 1/5 of the length from the left edge (-300 + 600/5 = -180)
circle_center_x = -300 + (length / 5)
radius = 80

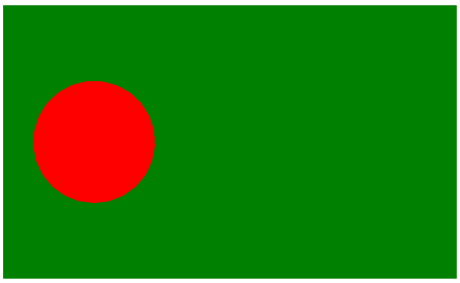
penup()
goto(circle_center_x, -radius)
pendown()
color("red", "red")
begin_fill()
circle(radius)
end_fill()

hideturtle()
done()
```

Problem Description: Draw a green background rectangle with proportions of approximately 10:6 (length to height). Draw a solid red circle inside the rectangle. Position the centre of the red circle at one-fifth (1/5) of the rectangle's length from the left edge.

Methodology: The flag is parameterized with length 600 and height 360, drawn from $(-300, -180)$ to center it at origin $(0, 0)$. The red circle (radius 80) is translated horizontally to center $X_c = -300 + (600/5) = -180$, and filled using `begin_fill()` and `end_fill()`.

Output Visualization:



Result & Observation: In the authentic National Flag of Bangladesh, the circle is located at 9/20 (45%) from the left edge. Here, placing it at 1/5 (20%) shifts the red disc towards the left hoist border, strictly meeting the experimental specification.

Task 3: Overlapping Concentric Squares (8-Point Star)

GitHub Repository Link:
https://github.com/Masud744/Computer-Graphics-Lab/blob/main/task3.py

Python Source Code (`task3.py`):

```python
import math
from turtle import *

speed(3)
pensize(2)

size = 200

# 1. First square (straight/upright)
penup()
goto(-size / 2, -size / 2)
pendown()
for i in range(4):
    forward(size)
    left(90)

# 2. Second square (rotated by 45 degrees - diamond shape)
# Distance from center (0,0) to corner vertex along axes = (size / 2) * sqrt(2)
diagonal_offset = (size / 2) * math.sqrt(2)

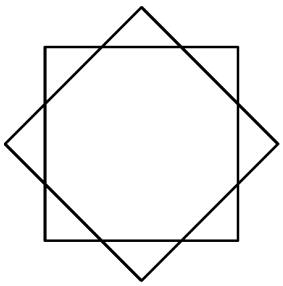
penup()
goto(0, -diagonal_offset)
setheading(45)
pendown()
for i in range(4):
    forward(size)
    left(90)

hideturtle()
done()
```

Problem Description: Draw the shape consisting of two overlapping squares sharing the identical center point $(0, 0)$. One square is aligned horizontally and vertically $(0°)$, and the second square is rotated by $45°$, forming an 8-pointed star shape.

Methodology: The upright square of side length $S = 200$ is drawn from corner $(-100, -100)$. The $45°$ rotated square is drawn with corner offset $R = \frac{S}{2}\sqrt{2} \approx 141.42$ along the vertical axis. Pen is set to $(0, -R)$ and directed via `setheading(45)` before drawing 4 equal sides.

Output Visualization:



Result & Observation: Both squares have identical side length $S = 200$ and center $(0, 0)$. Concentric intersection produces an 8-pointed regular star with flawless radial symmetry.

Task 4: Elegant Rotated Squares Pattern

GitHub Repository Link:
https://github.com/Masud744/Computer-Graphics-Lab/blob/main/task4.py

Python Source Code (`task4.py`):

```python
from turtle import *

speed(3)
pensize(2)

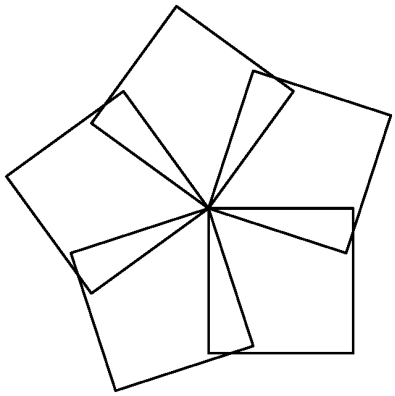
# Drawing 5 squares rotated around a single corner
# Total angle 360 / 5 = 72 degrees
for i in range(5):
    # Draw one square
    for j in range(4):
        forward(150)
        right(90)
    # Rotate 72 degrees for the next square
    right(72)

hideturtle()
done()
```

Problem Description: Draw the 5-square rotational pattern anchored at a common corner vertex in an elegant, concise, and efficient implementation.

Methodology: The pattern exhibits 5-fold rotational symmetry (C_5) about a common pivot vertex. It is implemented using a concise nested loop: repeating 5 times, turning right by $360°/5 = 72°$ after each 4-sided square with side 150 and $90°$ turns.

Output Visualization:



Result & Observation: By exploiting angular periodicity ($72°$), the entire 5-square pattern is generated concisely without repetitive code or coordinate hardcoding.